

The Cookie Countdown

A Rates Worksheet

October 31, 2026 · Nathaniel

Name: _____ Date: _____

The story

The bakery just baked **15,000 chocolate chip cookies**. That is a LOT of cookies!

Cookie Monster sneaks in. He eats **20 cookies every minute**. Nom nom nom.

The Count wants to count the cookies. But every time he looks up, there are fewer! *"Ah ah ah! The number keeps CHANGING!"*

Today we will help The Count. We will learn how to figure out how many cookies are left, even while Cookie Monster is still eating.

Big idea: a rate is "how much changes each minute"

Cookie Monster eats at the same speed the whole time: 20 cookies, then 20 more, then 20 more. That speed is called a **rate**.
The cookie pile goes **down** by 20 every minute. Mathematicians say the cookies are *changing* at a rate of **minus 20 cookies per minute**. The minus means "getting smaller."

Try it! Put 20 counters (beans, blocks, or paper cookies) in a pile. Take 20 away. That was one minute of Cookie Monster!

Question 1. Is the cookie pile getting bigger or smaller? _____

Question 2. How many cookies does Cookie Monster eat in 1 minute? _____

Warm-up: a small plate of 100 cookies

Before the giant bakery, let's practice with one plate of **100 cookies**. Cookie Monster still eats 20 every minute. Help The Count fill in the table by counting **back** by 20s.

Minutes gone by	Cookies eaten so far	Cookies left
0	0	100
1	20	80
2		
3		
4		
5		

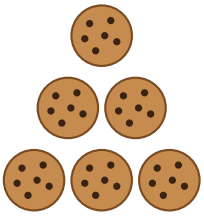
Question 3. After how many minutes is the plate empty? _____ minutes

Question 4. Look at the "Cookies left" column. How much smaller does it get each row? _____

That number is the **rate**. It is the same in every row! That is the secret The Count needs.

Related rates: the chocolate chips disappear too!

Every cookie has **10 chocolate chips**. When a cookie disappears, its 10 chips disappear with it. So the chips and the cookies are *related*: if we know one rate, we can find the other.



⟶ 10 chips gone

⟶ 20 chips gone

⟶ 30 chips gone

Count by 10s!

Question 5. Cookie Monster eats 20 cookies in one minute. How many chips is that? Count by 10s twenty times, or use 20 groups of 10. _____ chips per minute

Question 6. Which number goes down faster each minute: cookies or chips? _____

This is called **related rates**: two things change together, and one changes 10 times as fast as the other.

The big bakery: 15,000 cookies

15,000 is too many to count one at a time, even for The Count! So we use groups. From the warm-up, we know **one plate of 100 cookies lasts 5 minutes**.

Question 7. 10 plates make 1,000 cookies. If each plate takes 5 minutes, how long do 10 plates take? Count by 5s ten times.

plate	plate	plate	plate	plate	plate	plate	plate	plate	plate
____	____	____	____	____	____	____	____	____	____

_____ minutes

Question 8. The bakery has 15 groups of 1,000 cookies. Each group takes the answer from Question 7. Count by 50s fifteen times: 50, 100, 150, . . .

50	100	150					

_____ minutes

Question 9. That is a long time! It is about 12 and a half hours. If Cookie Monster starts at breakfast (7:00 in the morning), about when does he finish? Circle one:

Lunch time Dinner time Next week

Draw it!

Draw a big cookie pile. Draw an arrow pointing down and write "-20 every minute" next to it.

The Count's Calculus Club: write it like a scientist!

"Ah ah ah! You just did CALCULUS!" **Calculus** is the math of things that **change**. Scientists use it for rockets, rivers, and the Sun. Here is how they write what you already figured out.

Step 1: Give things short names

Scientists use one letter as a nickname.

C means the number of **C**ookies left

t means the **t**ime, in minutes

Step 2: The "change" letter, d

The letter d means **"a little change in."** So dC is a change in cookies, and dt is a change in time. Put them together:

$$\frac{dC}{dt} = -20$$

Say it out loud: **"dee-C dee-t equals minus twenty."** It means: *the cookies change by* -20 *every minute.* That is the **rate** you found in Question 4!

Step 3: A rule for the cookies left

For the plate of 100 cookies, the rule is

$$C = 100 - 20 \times t$$

It says: start with 100, then take away 20 for every minute.

Question 10. Use the rule to find the cookies left after $t = 2$ minutes. $100 - 20 \times 2 =$ _____
Does it match your warm-up table? _____

Step 4: Related rates, the scientist way

Each cookie has 10 chips. If P is the number of chi**P**s, then

$$\frac{dP}{dt} = 10 \times \frac{dC}{dt} = 10 \times (-20) = -200$$

The chips go down 200 every minute: your answer to Question 5! Grown-up calculus students call this the **chain rule**. You did it by counting by 10s.

Super challenge!

Now The Count counts **60 cookies every minute** from the pile, and Cookie Monster *still* eats 20 every minute from the same pile.

Question 11. How many cookies leave the pile each minute? $60 + 20 =$ _____

When two things take from the same pile, their **rates add**! A scientist would write

$$\frac{dC}{dt} = -20 - 60 = -80$$

With 15,000 cookies, the pile is empty after about 188 minutes: that is about 3 hours instead of 12 and a half!

For teachers: the Step 3 rule is the solution of the differential equation $dC/dt = -20$ with $C(0) = 100$; for the bakery, $C(t) = 15{,}000 - 20t$ reaches zero at $t = 750$ minutes (Question 8). In the challenge, $t = 15{,}000/80 = 187.5$ minutes, when The Count has counted 11,250 cookies and Cookie Monster has eaten 3,750.

Answer key

Question	Answer
1	Smaller
2	20 cookies
Warm-up table	Minute 2: 40 eaten, 60 left Minute 3: 60 eaten, 40 left Minute 4: 80 eaten, 20 left Minute 5: 100 eaten, 0 left
3	5 minutes
4	20 smaller each row (the rate)
5	200 chips per minute
6	Chips (10 times as fast)
7	50 minutes
8	750 minutes
9	Dinner time (about 7:30 in the evening)
10	60 cookies; yes, it matches minute 2 of the table
11	80 cookies per minute